Word Priming Experiment
PSY310 Lab in Psychology
Lab Report

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GitHub Link:
https://github.com/tamannaz06/Tamanna-Chandna_Word-Priming-Experiment

Introduction

Memory is often stored without consciously knowing its origin. It guides thought and behavior without requiring conscious recall. This type of memory is called implicit memory. One of the most common ways of studying this type of memory is priming.

Priming examines the subconscious effect of the stimulus that is first exposed that improves subsequent recall and task performance. When memory processes information unconsciously because of a short-term memory, we can say the priming effect has taken place. In the priming stage, schemas are linked to the presentation of the initial stimulus before being exposed to the next stimulus. These primed stimuli become more present to react to the later stimulus on the basis of the previous one. A variety of tasks have been used to assess priming effects or implicit learning, such as the word fragment task which is used in the current experiment.

It is a word fragment task in which some letters are "primed" for a particular word.
Some letters of a word are removed and the challenge is to say what the word was. The experiment investigates the number of times the participant is able to correctly complete the word fragment of the words from a priming phase as compared to the new words that were not introduced previously.

Using the word- fragment completion task, Tulving, Schacter and Stark (1982) showed that previously studied words were more likely to be correctly completed than words that had not been presented previously, thus demonstrating the relationship between implicit memory and the word-fragment completion task. Importantly, this priming effect was fairly consistent even when participants were no longer able to explicitly remember the previously presented words, indicating that implicit memory can be maintained without explicit or conscious recognition. This experiment aims to replicate this.
Further investigation of this process, on the other hand, revealed that the exact repetition of words supported word-fragment completion, while conceptual repetition (in the form of synonyms, associations and related words) did not (Roediger & Challis, 1992). This indicates that response to a word fragment is significantly dependent on perceptual or lexical knowledge of the words, rather than just on the activation of related representations. Roediger, Stadler, Weldon and Riegler (1992) also compared word-fragment and word-stem completion which is another method testing priming as measures of implicit memory. They found that both tasks could show priming effects and that such priming effects could last over time. All of these results indicate that word-fragment completion can be used as a measure of implicit memory.

Method

Participants

This experiment involves data from four undergraduate students (n=4) from various Indian Institutions. The age ranges from 18-23 (mean=20). Both male and females took part: 3 females

and 1 male. The demographics are shown in the table below. The experimenter used a convenience sampling technique to gather participants. Since the experiment was conducted during the Diwali break, participants were more diverse than usual but an age-range was kept. hours. Informed consent was received from each of the participants after they had understood the study's instructions. They were not told about the possibility of new words that had not been shown previously appearing in the test phase, to prevent demand characteristics as that would skew results. The participants' data was assigned numbers according to their order of participation to ensure confidentiality. No names or initials were saved.

Assigned no.	Age	Sex	Year of study
#1	18	Female	1st
#2	19	Female	1st
#3	23	Male	5th
#4	20	Female	3rd

Table 1: Details of the participants

Materials

The experiment was conducted on a laptop with PsychoPy Studio 2026.2.4 installed. The laptop has a monitor of dimension 13.3", a resolution of 1920x1080 pixels and a refresh rate of 60 Hz. 20 trials of the experiment were conducted. This is the same as the number of words tested. The aim was to see whether there is an improvement in performance for primed words in word fragment completion.

Stimuli and Procedure

The experiment had two phases: the study phase and the testing phase. Initially, as a part of the study phase, the participants were presented with 15 words to be read and familiarized with. Then the words were shown one by one and participants had to rate their familiarity with the word. This familiarity rating task was done with the help of a five-point rating scale slider which ranged from 'familiar' to 'unfamiliar' as shown in the figure below. The task was set in a way to prime the participants to the 15 familiar words before the next, testing phase of the experiment.

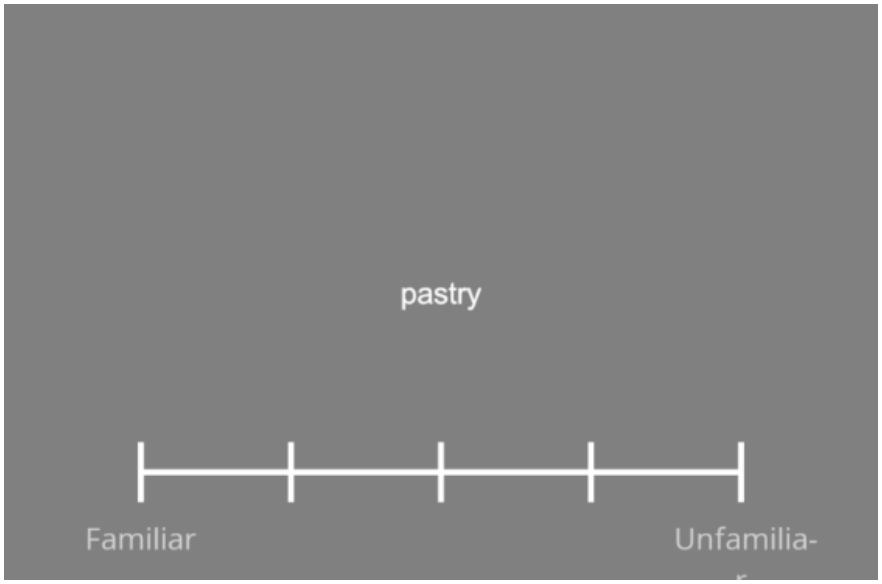


Figure 1: Familiarity rating task using slider in Study phase

Next, the test phase of the study was the part that assessed the priming effect. In this, participants performed a word fragment completion task. This involves a fill-in-the-blanks format of a word that has some letters removed and replaced with a '_'. This incomplete word appeared at the top of the screen and the participants had two tasks to perform once they saw that word. The first one was to type out the full word (not just the missing letters) and second was to click the assigned area at the bottom to submit their response. In this test phase, the 15 primed words appeared as well as 5 unprimed or new words.The order of these words was set to random. If the participant did not know the answer to some word, they were told to either submit a blank response, write any letters or take a guess. The experiment ended once all 20 words were responded to. Figures 2 and 3 show samples of word fragments for both primed and unprimed words.

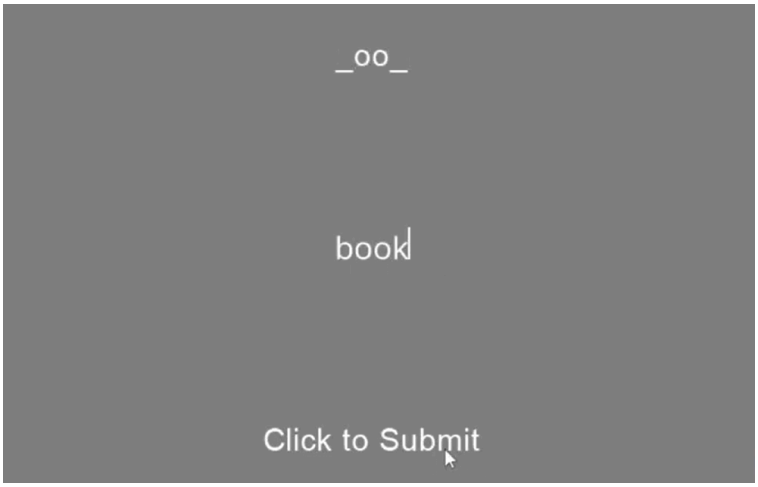


Figure 2: Phase 2- Word fragment with answer for a primed word"book"



Figure 3: Phase 2- Word fragment for an unprimed word“vase”

testwords	***reference***	***old_new***
l_pt_p	*Laptop*	*old*
_ha_t	*Chart*	*old*
T_hi_t	*Tshirt*	*old*
_ou_se	*mouse*	*old*
*P_str_*	*Pastry*	*old*
*c_s_*	*Case*	*old*
*_ruis_*	*Cruise*	*old*
P_li_e	*Police*	*old*
*K_y_*	*Keys*	*old*
P_st_r	*Poster*	*old*
_o_rd	*Board*	*old*
*_a_bl_*	*Marble*	*old*

testwords	*reference*	*old_new*
l_pt_p	*Laptop*	*old*
_ha_t	*Chart*	*old*
T_hi_t	*Tshirt*	*old*
_ou_se	*mouse*	*old*
_ho_e	*Phone*	*old*
*_oo_*	*Book*	*old*
*c_s_io_*	*Cushion*	*old*
*ta_l_*	*Table*	*new*
p_nc_l	*Pencil*	*new*
*v_s_*	*Vase*	*new*
bo_t_e	*Bottle*	*new*
c_m_ra	*Camera*	*new*

Table 2: Primed and unprimed word fragments (named testwords). These were the 20 words assessed.

For the purpose of this experiment, the analysis did not include familiarity ratings and reaction times as they were not relevant to the priming score. A hit was counted if participants had spelled out each letter of the word fragment accurately. However, If participants had not completed spelling out the word fragment, didn't know the word, left it blank or spelled it wrongly, it was considered a miss. If a participant generates their own word to complete the word fragments, but it was not a primed word, such a trial won't be considered.
The priming score is calculated by subtracting the proportion of correctly responded to words (hits) in primed trials from the proportion of hits in primed trials.
The formula is shown below:

$$Priming\ Score = \frac{No.\ of\ Hits\ from\ Primed\ List}{No.\ of\ Trials\ with\ Primed\ Words} - \frac{No.\ of\ Hits\ from\ Unprimed\ List}{No.\ of\ Trials\ with\ Unprimed\ Words}$$

This formula would be calculated through the excel sheet data. It will give the magnitude of the priming score. A score of 0 would mean that no priming took place, a positive score posits that priming has taken place and a negative score shows the opposite of priming, that is the primed words slowed down the recall of test words.

Results

Participant	No. of words trials excluded	Proportion of hits for primed words	Proportion of hits for unprimed words	Priming score
#1	0	0.666666667	0.8	-0.133333333
#2	0	0.8	0.5	0.3
#3	0	0.933333333	0.8	0.133333333
#4	0	0.8	0.8	0

Table 3: Summary of results from all 4 participants

Participant #1

- Proportion of hits from the list of primed words: 10/15 = 0.667
- Proportion of hits from the list of unprimed words: 4/5 = 0.8
- Priming score: 0.667 → 0.8 = −0.133

This participant completed the new words more often than the studied ones, so there is no sign of priming. There is, in fact, negative priming. This negative score is probably because the participant unconsciously suppressed the word when it was initially shown and could not access it during the test phase or perhaps didn't remember the words well enough initially.

Participant #2

- Proportion of hits from the list of primed words: 12/15 = 0.8
- Proportion of hits from the list of unprimed words: 2/5 = 0.4
- Priming score: 0.8 → 0.4 = 0.4

This is the clearest priming effect of the four. Studied words were completed twice as often as new or unprimed ones.

Participant #3

- Proportion of hits from the list of primed words: 14/15 = 0.933
- Proportion of hits from the list of unprimed words: 4/5 = 0.8
- Priming score: 0.933 → 0.8 = 0.133

Participant 3 shows a small priming effect. The participants only got 1 wrong in both the lists, but due to difference in denominator, the proportion changed and there was little room for the studied words to stand out.

Participant #4

- Proportion of hits from the list of primed words: 12/15 = 0.8
- Proportion of hits from the list of unprimed words: 4/5 = 0.8
- Priming score: 0.8 → 0.8 = 0

There is no priming effect. Studied and new words were completed equally often at 80%.

Overall, only participant 2 shows a clear effect, #3 a weak one and #1 and #4 show none. It's important to note that the new-word list has only five items, so one word changes a proportion by 0.2 which is a lot more compared to the proportional change by getting 1 word wrong in the test words.

Discussion

The expected priming effect did not appear strongly. 50% of the participants had a positive priming score while the remaining two participants had either a negative priming score or 0. Unfortunately, this performance is similar to chance levels and does not reveal much about the expected priming effect. Reasons for this could be the low number of unprimed words as well as participants. Also, the new words were reported to be quite easy to guess even without priming as the participants regularly write in english.

The total number of accurate trials per participant were 14,14,18 and 16 for participant number 1, 2,3 and 4 respectively. This is decent but not great, showing that perhaps the participants were not fully focused. Especially the first two.

It's also worth noting that the fragments of 'pastry' (p_str_) and 'poster' (p_st_r) confused people and they made mistakes there (3 out of 4 of them made this error) as they thought they were overlapping. This might be because the environment wasn't fully conducive for the experiment as it was done at home during the festive season.

Lastly, since priming is an unconscious encoding of information, the experimenter cannot share too many details about the experiment with participants beforehand as it would make it explicit and defeat the purpose. These factors put together try to explain the results.

It’s possible to study this priming effect using faces as a stimuli too since faces are closer to the way we process information in daily life. The idea that people trust what feels familiar plays out in many settings. It’s seen in simple settings like a shopper may be swayed by a celebrity in an advertisement and in more serious cases like a jury’s view of a defendant may be shaped by how familiar or trustworthy that person seems which can be catastrophic.

For this experiment, I would definitely aim to have more participants. At least 30-40 with an equal distribution of males and females. It will be within-subject design and informed consent would be taken as always.

One way to test facial priming that comes to my mind right now is to see whether faces that participants have seen before are judged faster. In the first phase, participants can look at 30 faces, one at a time for a few seconds each and say whether each person looks friendly or not. This keeps their attention on the faces without revealing the actual purpose of the study.

In the second (test) phase, they will see 50 faces: 30 from the study phase and 20 new ones. For each face, they press one key for “male” and another for “female” as quickly as they can. This task has nothing to do with remembering, so any advantage for the earlier faces would come from priming and not from deliberate recall.

If the earlier faces get quicker responses than the new ones, priming has occurred.

References

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